



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

(An Autonomous Institution Affiliated to Anna University, Chennai)

SATHYAMANGALAM - 638 401

Module Test - II - May - 2026

II Semester

22GE004 & BASICS OF ELECTRONICS ENGINEERING

1	(i)	During testing of a digital voting system, engineers observe that certain input combinations produce incorrect outputs due to suspected misinterpretation of logic levels and possible inversion errors in signal processing, where binary inputs correspond to voter validation signals and the expected outputs are predefined using Boolean expressions derived earlier in the design phase. Diagnose the possible logical inconsistencies and determine corrective measures by interpreting the relationship between binary inputs, logic levels, and Boolean representation (3 Marks - [An/C, 2])
2	(ii)	A digital signal processor receives two binary inputs representing encoded sensor signals, and the system must mathematically model the logical relationship using Boolean expressions to ensure that the output satisfies a specified truth condition, while also enabling simplification for efficient hardware realization. Formulate and manipulate the Boolean expression to achieve a simplified logical representation that satisfies the given conditions. (2 Marks - [Ap/C, 2])
3	(i)	A control module must implement both AND and OR operations using only diodes due to fabrication limits. Multiple stages are required, but signal degradation is observed across stages. Engineers must design a stable configuration while maintaining simplicity. The absence of amplification increases complexity in maintaining logic levels. Integrate diode behavior and circuit principles to construct a multi-stage logic system and justify how signal degradation is managed across stages. (3 Marks - [U/C, 2])
4	(ii)	A consumer device requires low-cost logic implementation and considers diode logic. Engineers observe signal degradation and environmental sensitivity. The system must operate reliably over time. A professional recommendation is required. Assess the suitability of diode logic and recommend its adoption considering performance and reliability constraints. (2 Marks - [An/C, 2])
5	(i)	A robotics control unit processes binary inputs from multiple switches representing operational modes, and the system must follow a structured procedure to convert these inputs into logical decisions using standard logic gates, ensuring that each stage of processing aligns with Boolean algebra rules and produces deterministic outputs under all input conditions. Determine the procedural flow required to transform binary inputs into final logical outputs while maintaining consistency with Boolean laws and gate-level implementation. (3 Marks - [Ap/C, 2])

6	(ii)	A digital systems engineer is reviewing a circuit design for compliance with industry standards where logic gates are used to implement safety-critical operations, and the evaluation must consider correctness of Boolean expressions, proper interpretation of binary logic levels, and robustness of the overall design under varying electrical conditions. Assess the design reliability by integrating logical correctness, representation accuracy, and practical implementation considerations. (2 Marks - [An/C, 2])
7	(i)	A technician builds a diode logic circuit implementing OR followed by AND operations. Proper diode orientation and resistor placement are critical. Incorrect connections lead to faulty outputs. The setup must be reproducible in a lab environment. Formulate a structured procedure to construct the circuit and justify each step based on logic and circuit behavior. (3 Marks - [Ap/C, 2])
8	(ii)	A system integrates multiple inputs using diode logic, but output voltage decreases as inputs increase. Reliability issues arise due to loading effects. The designer must evaluate scalability limits and maintain performance. Analyze scalability limitations of diode logic and prioritize trade-offs to maintain efficiency. (2 Marks - [An/C, 2])
9	(i)	A digital signal conditioning circuit uses a PN junction diode where input voltage gradually increases from 0 V to 0.9 V, and the designer must determine the exact transition point where the diode switches from non-conducting to conducting state, considering the exponential V-I relationship and the presence of a threshold voltage around 0.7 V that governs switching behavior in digital applications. Formulate the switching condition using the diode V-I characteristics and analyze how the exponential relationship governs the transition between OFF and ON states for accurate signal interpretation. (5 Marks - [Ap/C, 2])
10	(i)	A low-cost embedded monitoring device is being developed for a remote agricultural system where environmental noise fluctuates due to nearby motor operations and long wiring paths between sensors and control units. The design team initially considers using resistor-transistor logic to implement control signals but observes inconsistent switching when signal levels degrade over distance. Another engineer proposes replacing part of the design with diode-transistor logic to improve robustness while maintaining cost constraints. The system must operate reliably under moderate noise and limited power availability, and must also ensure consistent switching thresholds despite temperature variations. The team must decide which logic family to adopt for critical signal paths while maintaining simplicity in non-critical sections. Analyze the given system constraints and determine an optimal allocation strategy between RTL and DTL logic families for different parts of the circuit, justifying your decision based on operational reliability, noise handling, and switching behavior. (5 Marks - [An/C, 2])
11	(i)	An industrial control circuit uses a PN junction diode for switching between sensor outputs and a logic controller, where engineers observe delayed switching and unintended signal overlap during high-frequency operation, and measurements indicate increased reverse recovery time and elevated leakage current under thermal stress, leading to incorrect logic interpretation and unreliable system performance in critical automation tasks. Analyze the underlying causes of malfunction by correlating reverse recovery time and leakage current with observed switching errors and recommend corrective strategies to restore proper functionality. (5 Marks - [An/C, 2])

12	(i)	A digital control board in an industrial automation unit begins to exhibit intermittent logic failures in a NOR gate implemented using resistor-transistor logic. The output occasionally remains in an undefined state when multiple inputs transition simultaneously. Measurements show that the transistor does not fully saturate under certain input combinations, and voltage drops across resistors vary with load conditions. The circuit operates in a high-temperature environment, which further affects transistor characteristics. Engineers suspect that improper current flow and insufficient drive are causing the malfunction, leading to unreliable switching behavior. Diagnose the root causes of the malfunction in the RTL NOR gate and infer how circuit-level limitations contribute to the observed instability, proposing corrective design interventions. (5 Marks - [An/C, 2])
13	(i)	A BJT switch operates with 10 V supply and 400 Ω load, with saturation voltage of 0.2 V. The engineer uses load line analysis to determine operating limits. The system must confirm saturation during ON state. Base current is adjusted accordingly. Determine the saturation collector current using load line analysis and infer whether the transistor operates correctly as a switch. (5 Marks - [Ap/C, 2])
14	(i)	A digital circuit designer is optimizing a Boolean function representing a safety monitoring system in an industrial plant where multiple sensors trigger alarms based on combined conditions. The original logic expression contains redundant terms leading to unnecessary gate usage and increased propagation delay. The designer aims to minimize the logic expression using Boolean algebra to reduce hardware complexity and improve speed while maintaining functional correctness. The minimized circuit must also be compatible with transistor logic implementation for integration into an IC. Formulate the process of minimizing the Boolean function and analytically justify how reduction in logic levels improves propagation delay and hardware efficiency within the given system. (5 Marks - [Ap/C, 2])
15	(i)	A safety indicator circuit uses a BJT switch, but the LED glows dimly when OFF. The base-emitter voltage is 0.25 V and ambient temperature is high. The circuit lacks a base-emitter resistor. The system requires complete cutoff. Analyze the cause of improper cutoff operation and recommend corrective measures to achieve complete OFF state switching. (5 Marks - [An/C, 2])
16	(i)	A semiconductor design firm is developing a compact control unit for an autonomous irrigation system deployed in remote agricultural fields where power supply is unstable and environmental noise is high due to long wiring and electromagnetic interference from pumps. The engineers must choose between diode logic, resistor-transistor logic, diode-transistor logic, and improved transistor logic families while ensuring reliable operation under fluctuating voltages. The system requires moderate switching speed, minimal signal degradation, and long operational life with limited maintenance access. Cost constraints are moderate but system failure must be avoided at all times due to crop dependency. Determine the most appropriate logic family for the described system by evaluating operational constraints and performance parameters. Justify the selection by integrating considerations of noise margin, power consumption, and scalability within the given scenario. (5 Marks - [Ap/C, 2])

17	(i)	A researcher is attempting to design a hybrid logic circuit that combines diode-based OR functionality with transistor-based inversion to create a compact logic unit. The system must maintain logical correctness while minimizing component count. However, the interaction between diode voltage drops and transistor threshold requirements complicates the design. The researcher must ensure that output levels from the diode stage are sufficient to drive the transistor stage effectively. Environmental variations further influence voltage thresholds and switching behavior. Integrate the functional roles of diodes and transistors to formulate a coherent logic design strategy that ensures reliable operation under given constraints. (3 Marks - [U/C, 2])
18	(ii)	A design review board evaluates multiple digital logic implementations for a commercial product. The team must ensure that the chosen logic design meets reliability, cost, maintainability, and future upgrade requirements. The product will operate in diverse environments, requiring consistent performance. Engineers must justify their design choice considering both technical and professional constraints. Assess the design decision criteria and recommend a balanced approach that aligns technical performance with professional engineering standards. (2 Marks - [U/C, 1])
19	(i)	An electronics researcher studies semiconductor capacitance mechanisms to optimize high-frequency tuning circuits requiring rapid response and minimal delay. The system operates under reverse bias and compares transition capacitance from depletion region variation with diffusion capacitance associated with charge storage. It is observed that one mechanism supports instantaneous response while the other introduces delay due to stored carriers. The researcher must identify the appropriate mechanism for efficient tuning. Integrate the characteristics of semiconductor capacitance mechanisms and infer their suitability for high-frequency tuning by linking physical behavior with system requirements. (3 Marks - [U/C, 2])
20	(ii)	A consumer electronics firm evaluates tuning mechanisms for compact communication devices where size and reliability are critical. Mechanical capacitors are compared with semiconductor-based voltage-controlled capacitors considering response speed and durability. Engineers note that mechanical systems degrade over time while semiconductor devices offer faster response but introduce nonlinear behavior. The firm must decide the most suitable approach for mass production. Assess semiconductor-based tuning devices compared to mechanical alternatives by prioritizing performance and manufacturing considerations and justify the selection. (2 Marks - [U/C, 1])
21	(i)	An engineer is tasked with constructing a digital switching circuit using BJTs operating as switches. The design must ensure clear transitions between ON and OFF states while minimizing switching delay. The circuit must operate within cutoff and saturation regions only, avoiding the active region. The engineer must also consider timing parameters such as rise time and fall time to ensure proper synchronization with other circuit components. Load conditions vary, requiring stable switching performance. Formulate a logical sequence of operational steps to ensure effective transistor switching while maintaining timing and stability requirements. (3 Marks - [U/C, 2])

22	(ii)	A circuit designer analyzes a PN junction diode used as a switch where the forward voltage drop is 0.7V and supply voltage is 5V. The diode is connected in series with a resistor, and the output is taken across the diode. The designer needs to determine whether the diode operates effectively as a switch under given conditions. Variations in input voltage may affect conduction behavior. Determine the operational state of the diode and infer its effectiveness as a switching element under the given electrical conditions. (2 Marks - [U/C, 1])
23	(i)	A circuit designer is implementing a voltage-controlled oscillator using a semiconductor-based capacitance element. The design involves establishing reverse bias conditions, integrating the device into an LC tank circuit, and ensuring stable oscillations across a defined frequency range. The system must respond linearly to control voltage while minimizing distortion caused by parasitic elements. Proper sequence of design steps is essential for effective implementation. Formulate a logical procedure for implementing the oscillator using a voltage-dependent capacitance device by structuring the sequence of design and integration steps. (3 Marks - [U/C, 2])
24	(ii)	A distributed communication network requires efficient tuning across multiple nodes operating under varying environmental conditions. Engineers consider voltage-controlled capacitive devices for scalability and low power consumption. However, nonlinear behavior and temperature sensitivity must be addressed to ensure consistent performance across all nodes. Analyze trade-offs in implementing voltage-controlled capacitive tuning across the network and recommend strategies to improve efficiency and scalability. (2 Marks - [U/C, 1])
25	(i)	A portable medical device designed to operate at 5.6 V begins to show unstable output voltage after prolonged usage. The input supply remains constant at 12 V, but the output fluctuates between 4.8 V and 6.2 V. The circuit uses a Zener diode regulator, and measurements reveal that the Zener current drops significantly when load demand increases. Further inspection shows that the series resistor value may have been altered during maintenance. The device occasionally shuts down, indicating intermittent loss of regulation, especially when load current peaks. Analyze the circuit behavior to identify the root cause of voltage instability and infer how improper component selection affects Zener regulation under varying load conditions. (5 Marks - [Ap/P, 3])
26	(i)	An LED is powered using a 5 V supply and operates at 2 V with desired current of 20 mA; a resistor is used to limit current; temperature variation may slightly affect forward voltage; the engineer must ensure safe operation while minimizing power loss; analytical calculation is required. Formulate and compute the appropriate current-limiting resistance and analyze the power distribution in the circuit under nominal conditions, integrating V-I characteristics and efficiency considerations. (5 Marks - [Ap/P, 3])

27	(i)	An industrial sensor module operates from an unstable DC supply that fluctuates between 14 V and 20 V due to varying load conditions in a manufacturing line. The sensor requires a stable 10 V supply for accurate measurements, and any deviation beyond ± 0.5 V results in incorrect readings. A Zener diode with a nominal breakdown voltage of 10 V is available, but its maximum power dissipation is limited. The load resistance varies dynamically due to switching sensor states, and improper resistor selection may either push the Zener into non-regulating mode or cause excessive current. The engineer must ensure reliable voltage regulation under all operating conditions while preventing component damage and maintaining efficiency. Formulate an optimal design strategy for the Zener diode regulator circuit by determining appropriate resistance and operating conditions, and justify how the design ensures consistent regulation under both supply and load variations. (5 Marks - [Ap/P, 3])
28	(i)	A wearable biomedical device uses an LED for optical sensing and must operate with low power and minimal heat generation; engineers must select a semiconductor material ensuring suitable wavelength for tissue penetration and stable emission; constraints include limited battery life and compact size; selection must balance efficiency, reliability, and thermal safety. Determine the most appropriate LED material and emission type for the described wearable system by integrating bandgap considerations, efficiency constraints, and application requirements, and justify the selection through structured reasoning linking device physics and operational constraints. (5 Marks - [Ap/P, 3])
29	(i)	A telecommunications backup system initially uses a Zener diode regulator for generating a stable reference voltage. As the system expands, additional modules increase load demand and introduce higher ripple in the input supply. Engineers consider replacing the Zener regulator with an integrated voltage regulator IC. The Zener circuit exhibits increased power dissipation and poor ripple rejection under scaling conditions. However, it is simple and low-cost. The integrated regulator offers better efficiency and stability but adds complexity and cost. The design team must evaluate scalability, thermal performance, and regulation quality under expanding system requirements. Evaluate the trade-offs between Zener-based and integrated voltage regulators under increasing load and ripple conditions, and determine the most suitable choice for scalable system deployment. (5 Marks - [Ap/P, 3])
30	(i)	A solar system shows reduced output despite good sunlight. Some panels exhibit hotspots and irregular current flow. Dust and partial shading are present, and panels of different types are used. Engineers must identify causes of performance loss. Analyze the scenario to diagnose the underlying causes of performance degradation and recommend targeted interventions based on photovoltaic principles and equivalent circuit behavior. (5 Marks - [Ap/P, 3])
31	(i)	A laboratory power supply using a Zener diode regulator begins to produce unstable output voltage after prolonged operation. Measurements show that output voltage drifts upward as temperature increases. The Zener diode used has a positive temperature coefficient. The circuit operates close to maximum power dissipation limits. Engineers suspect thermal instability affecting regulation accuracy. The system lacks adequate heat sinking and operates continuously in a high ambient temperature environment. The goal is to identify the root cause and propose corrective measures to stabilize the output. Diagnose the cause of voltage instability in the regulator and determine corrective actions based on thermal characteristics and safe operating conditions. (5 Marks - [Ap/P, 3])

32	(i)	Scenario: A rural electrification project plans a solar microgrid in an area with seasonal shading and high temperatures. Energy demand varies between daytime irrigation and nighttime lighting. Budget constraints limit cooling solutions, and dust accumulation affects performance. The system must integrate storage for reliability. Analyze the given scenario and formulate an optimized design strategy for selecting solar cell type, arrangement, and operational considerations that maximize energy conversion efficiency and system reliability under the stated constraints, while justifying trade-offs involved. (5 Marks - [Ap/P, 3])
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